

POWER ANALYSERS Becoming Portable And Power-Efficient



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The shift to handheld equipment has been going on for a long time. We now have handheld power supplies, multimeters and thermal imagers, among others. This is just one of the many changes test and measurement (T&M) equipment have undergone over the past few years.

Low power consumption is another. In the process of developing and testing such equipment, power analysers have been very useful. So in this issue, we take a look at some of their latest features.

Custom logging is just one of the features

Traditional power meters were discrete products. "Those were designed to test basic power parameters at a nominal accuracy. Most of these were designed for testing active average power and did not have high enough sampling rates or bandwidth to handle modern non-linear power-converter signals," says Nitin Nigam, application engineer, Tektronix.

Compared to those, recent power analysers offer much more. Today we have "custom logging features, waveform,

harmonics and trend graphs to address all modern power analysis needs," adds Nigam.

Power analysers offer insulated current input, measurement of harmonics, alternating and direct current (AC/DC), voltage, frequency, phase-rotation, and idle, apparent and effective power measurements. Basically you get almost all the power measurements of the device under test from a single piece of equipment, which is very convenient.

Flexibility of testing single-, two- and three-phase applications has also been an area of focus. You can perform testing in three-wire two-wattmeter method, four-wire three-wattmeter method or two-wire one-wattmeter method. Integrated software enhances application and flexibility of use, too.

These equipment can measure accurately even at low power factors such as 0.1 with up to 0.1 per cent basic accuracy. Usage of 16-bit microcontrollers and 24-bit analogue-to-digital converters allows accuracy and calculation of voltage, current, power, power factor, crest factor, form factor and frequency.

